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Assignment 4

1. Give a FOL translations for the following sentences: (10M) (CO4) (BLT4)
   a. Vegetarians do not eat meat.
   b. Not all vegetarians eat eggs.

2. Give a set of facts and inferences necessary to prove the following assertions: (10M) (CO4) (BLT4)
   a. McDonalds is not a vegetarian restaurant.
   b. Some vegetarians can eat at McDonalds.

Don't just place these facts in your knowledge base. Show that they can be inferred from some more general facts about vegetarians and McDonalds.

3. Give FOL translations for the following sentences that capture the temporal relationships between the events. (10M) (CO4) (BLT4)
   a. When Mary's flight departed, I ate lunch.
   b. When Mary's flight departed, I had eaten lunch.

4. The following rule, which we gave as a translation, is not a reasonable definition of what it means to be a vegetarian restaurant. (10M) (CO4) (BLT4)

$\forall x \, VegetarianRestaurant(x) \Rightarrow Serves(x, VegetarianFood)$

Answer 1. FOL translations

a. Vegetarians do not eat meat.

$\forall x \, \forall y \, (Vegetarian(x) \wedge Meat(y) \Rightarrow \neg Eats(x,y))$

Event form: $\forall x \, (Vegetarian(x) \Rightarrow \neg \exists e \, \exists y \, (Eating(e) \wedge Eater(e,x) \wedge Eaten(e,y) \wedge Meat(y)))$

Every vegetarian x and every piece of meat y are such that x does not eat y; equivalently, there is no eating event in which a vegetarian eats meat.

b. Not all vegetarians eat eggs.

$\neg \forall x \, (Vegetarian(x) \Rightarrow \exists y \, (Egg(y) \wedge Eats(x,y)))$

Equivalent: $\exists x \, (Vegetarian(x) \wedge \neg \exists y \, (Egg(y) \wedge Eats(x,y)))$

Pushing the negation inward ($\neg\forall$ becomes $\exists\neg$, and $\neg(A \Rightarrow B)$ becomes $A \wedge \neg B$) shows that there is at least one vegetarian who eats no eggs. This is weaker than "no vegetarian eats eggs", which would be a universal negation.

Answer 2. Facts and inferences

Instead of asserting the conclusions, a few general facts about vegetarians, food and restaurants are placed in the knowledge base, and both assertions are derived from them.

General facts (rules):

R1 $\forall x \, (VegetarianRestaurant(x) \Rightarrow Restaurant(x) \wedge \forall y \, (Serves(x,y) \Rightarrow VegetarianFood(y)))$

R2 $\forall y \, (VegetarianFood(y) \Leftrightarrow \neg ContainsMeat(y))$

R3 $\forall y \, (Hamburger(y) \Rightarrow ContainsMeat(y))$

R4 $\forall y \, (Salad(y) \Rightarrow \neg ContainsMeat(y))$

R5 $\forall x \, \forall y \, (Vegetarian(x) \wedge VegetarianFood(y) \Rightarrow CanEat(x,y))$

R6 $\forall x \, \forall r \, \forall y \, (Serves(r,y) \wedge CanEat(x,y) \Rightarrow CanEatAt(x,r))$

Specific facts about McDonalds:

F1 $Restaurant(McDonalds)$

F2 $Serves(McDonalds, BigMac)$

F3 $Hamburger(BigMac)$

F4 $Serves(McDonalds, SideSalad)$

F5 $Salad(SideSalad)$

F6 $Vegetarian(Mary)$

a. McDonalds is not a vegetarian restaurant.

#	Statement	Justification
1	$Hamburger(BigMac)$	F3
2	$ContainsMeat(BigMac)$	R3, universal instantiation + modus ponens with 1
3	$\neg VegetarianFood(BigMac)$	R2 ($\Leftarrow$ read left to right in contrapositive) with 2
4	$Serves(McDonalds, BigMac)$	F2

#	Statement	Justification
5	*Serves*(*McDonalds*,*BigMac*) ∧ ¬*VegetarianFood*(*BigMac*)	∧-introduction, 3 and 4
6	¬∀*y* (*Serves*(*McDonalds*,*y*) ⇒ *VegetarianFood*(*y*))	from 5 (a counterexample y = BigMac)
7	*VegetarianRestaurant*(*McDonalds*) ⇒ ∀*y* (*Serves*(*McDonalds*,*y*) ⇒ *VegetarianFood*(*y*))	R1, universal instantiation, ∧-elimination
8	¬*VegetarianRestaurant*(*McDonalds*)	modus tollens, 6 and 7

b. Some vegetarians can eat at McDonalds.

#	Statement	Justification
1	*Salad*(*SideSalad*)	F5
2	¬*ContainsMeat*(*SideSalad*)	R4 with 1 (modus ponens)
3	*VegetarianFood*(*SideSalad*)	R2 (⇐ direction) with 2
4	*Vegetarian*(*Mary*) ∧ *VegetarianFood*(*SideSalad*)	F6, 3, ∧-introduction
5	*CanEat*(*Mary*, *SideSalad*)	R5 with 4 (modus ponens)
6	*Serves*(*McDonalds*, *SideSalad*) ∧ *CanEat*(*Mary*, *SideSalad*)	F4, 5, ∧-introduction
7	*CanEatAt*(*Mary*, *McDonalds*)	R6 with 6 (modus ponens)
8	∃*x* (*Vegetarian*(*x*) ∧ *CanEatAt*(*x*, *McDonalds*))	F6, 7, existential generalisation (x = Mary)

Neither conclusion was stored directly. (a) follows from general facts about meat, hamburgers and what a vegetarian restaurant must serve, and (b) follows from the facts that vegetarians can eat vegetarian food and that a person can eat at any place that serves something they can eat.

Answer 3. Temporal relationships between events

Each sentence is represented with event variables and explicit time intervals (Reichenbach-style). Eating(e) and Departing(e) are event types, i_e is the interval of event e, Now is the time of speaking, Precedes(i,j) means interval i ends before j begins, and Overlaps(i,j) means the two intervals coincide in time.

a. When Mary's flight departed, I ate lunch.

∃*e1* ∃*e2* ∃*f* ∃*l* ∃*i* ∃*j* (*Flight*(*f*) ∧ *FlightOf*(*f*,*Mary*) ∧ *Departing*(*e1*) ∧ *Departer*(*e1*,*f*) ∧ *IntervalOf*(*e1*,*i*) ∧ *Lunch*(*l*) ∧ *Eating*(*e2*) ∧ *Eater*(*e2*,*Speaker*) ∧ *Eaten*(*e2*,*l*) ∧ *IntervalOf*(*e2*,*j*) ∧ *Overlaps*(*i*,*j*) ∧ *Precedes*(*i*,*Now*) ∧ *Precedes*(*j*,*Now*))

Both events are in the past (simple past), and the eating interval j coincides with the departure interval i: the speaker was eating lunch at the time the flight departed.

b. When Mary's flight departed, I had eaten lunch.

∃*e1* ∃*e2* ∃*f* ∃*l* ∃*i* ∃*j* (*Flight*(*f*) ∧ *FlightOf*(*f*,*Mary*) ∧ *Departing*(*e1*) ∧ *Departer*(*e1*,*f*) ∧ *IntervalOf*(*e1*,*i*) ∧ *Lunch*(*l*) ∧ *Eating*(*e2*) ∧ *Eater*(*e2*,*Speaker*) ∧ *Eaten*(*e2*,*l*) ∧ *IntervalOf*(*e2*,*j*) ∧ *Precedes*(*j*,*i*) ∧ *Precedes*(*i*,*Now*))

The past perfect ("had eaten") places the eating interval j completely *before* the departure interval i, and the departure itself is in the past. In Reichenbach's terms, (a) has event time E = reference time R, whereas (b) has E < R < speech time S. The only difference between the two formulas is Overlaps(i,j) versus Precedes(j,i).

Answer 4. Why the rule is not a good definition, and a better one

Rule given: $\forall x\ VegetarianRestaurant(x) \Rightarrow Serves(x, VegetarianFood)$

Problems with the rule:

1. It is only a one-way (necessary) condition. It says a vegetarian restaurant serves vegetarian food, but a restaurant that serves nothing but vegetarian food cannot be proved to be a vegetarian restaurant from it.
2. It is too weak. “Serves vegetarian food” is satisfied by any restaurant with a single vegetarian dish, so McDonalds (which serves a salad) would pass, although it also serves hamburgers. A vegetarian restaurant should serve *only* vegetarian food.
3. VegetarianFood is used as a single object (a constant), not as a category of dishes, so the rule cannot talk about individual dishes such as a particular salad or burger.
4. It does not say that the thing is a restaurant at all.

Better definition: a vegetarian restaurant is a restaurant that serves something, and everything it serves is vegetarian food.

$$\forall x\ (VegetarianRestaurant(x) \Leftrightarrow Restaurant(x) \wedge \exists y\ Serves(x,y) \wedge \forall y\ (Serves(x,y) \Rightarrow VegetarianFood(y)))$$

This is a two-way definition ($\Leftrightarrow$), so it both lets us conclude that a restaurant is vegetarian when all its dishes are vegetarian and lets us rule it out when it serves even one non-vegetarian dish, as in Answer 2(a). The clause $\exists$y Serves(x,y) stops an empty restaurant that serves nothing from counting vacuously as vegetarian.

Supplementary Explanation - Answer 1: First-Order Logic

The first question translates ordinary-language statements into formulas that describe people, objects, and relations. A formula is useful only when its predicates and quantifiers express the same scope as the original sentence.

1.1 Predicates and variables used

Symbol	Meaning in this assignment
Vegetarian(x)	x is a person who is vegetarian.
Meat(y)	y is an item of meat.
Egg(y)	y is an egg.
Eats(x,y)	person x eats item y.
Eating(e), Eater(e,x), Eaten(e,y)	an eating event e, its eater, and the item eaten.

The variables x and y range over the entities in the chosen domain. The predicates describe properties or relations; they are not English words inserted into a formula. For example, Eats(x,y) is a two-place relation because it connects an eater with something eaten.

1.2 “Vegetarians do not eat meat”

The sentence makes a general claim about every vegetarian and every meat item. The universal quantifiers therefore appear before the implication:

$$\forall x\, \forall y\, (Vegetarian(x) \land Meat(y) \Rightarrow \neg Eats(x,y))$$

The antecedent Vegetarian(x) ∧ Meat(y) identifies the relevant pair: x must be vegetarian and y must be meat. The consequent ¬Eats(x,y) states that this pair is not related by Eats. If either part of the antecedent is false, the rule does not make a claim about that pair.

The same idea can be written by ruling out a forbidden situation:

$$\neg\exists x\, \exists y\, (Vegetarian(x) \land Meat(y) \land Eats(x,y))$$

This alternative says that there is no pair consisting of a vegetarian and a meat item such that the vegetarian eats that item. It is equivalent to the universal implication above: both prohibit a vegetarian-eats-meat case.

The assignment also gives an event-based version. It adds an event variable e and states that no eating event has a vegetarian as its eater and a meat item as its object:

$$\forall x\, (Vegetarian(x) \Rightarrow \neg\exists e\, \exists y\, (Eating(e) \land Eater(e,x) \land Eaten(e,y) \land Meat(y)))$$

The event representation is more detailed because it describes an occurrence of eating. The simpler Eats(x,y) representation treats eating as a direct relation. Both are ways of expressing the same central restriction in this question.

Supplementary Explanation - Answer 1 (continued)

1.3 “Not all vegetarians eat eggs”

This statement is easy to confuse with “no vegetarians eat eggs.” The phrase “not all” negates a universal statement. It does not claim that every vegetarian avoids eggs.

The formula from the answer is:

$$\neg\forall x\ (Vegetarian(x) \Rightarrow \exists y\ (Egg(y) \wedge Eats(x,y)))$$

Read the inner part first. For a person x, ∃y (Egg(y) ∧ Eats(x,y)) says that x eats at least one egg. The implication says that every vegetarian has this property. The outer negation says that this universal claim is false.

Applying the standard quantifier-negation rules gives:

$$\neg\forall x\ A(x) \equiv \exists x\ \neg A(x)$$

$$\neg(Vegetarian(x) \Rightarrow B(x)) \equiv Vegetarian(x) \wedge \neg B(x)$$

Therefore the sentence is equivalent to:

$$\exists x\ (Vegetarian(x) \wedge \neg\exists y\ (Egg(y) \wedge Eats(x,y)))$$

The formula identifies at least one vegetarian who does not eat any egg. The negation is inside the existential statement about y, so it rules out every egg as an item eaten by that particular vegetarian.

1.4 Comparing similar English statements

English statement	Logical form	What it commits us to
Not all vegetarians eat eggs.	¬∀x (V(x) ⇒ ∃y (Egg(y) ∧ Eats(x,y)))	At least one vegetarian eats no eggs.
No vegetarians eat eggs.	∀x (V(x) ⇒ ¬∃y (Egg(y) ∧ Eats(x,y)))	Every vegetarian eats no eggs.
Some vegetarians eat eggs.	∃x (V(x) ∧ ∃y (Egg(y) ∧ Eats(x,y)))	At least one vegetarian eats an egg.

A simple way to test the first statement is to imagine two vegetarians: one eats eggs and the other does not. In that situation, “not all vegetarians eat eggs” is true, while “no vegetarians eat eggs” is false. This example shows why the position of the negation matters.

1.5 Scope is the key idea

A quantifier controls the variable that follows it within its scope. Moving a negation across a universal quantifier changes ∀ to ∃, and moving it across an existential quantifier changes ∃ to ∀. These transformations are not optional stylistic changes: they determine whether a statement is about every member of a group or at least one member.

For the two sentences in Question 1, the first is universal and restrictive; the second is existential after negation and requires only one counterexample to the claim that all vegetarians eat eggs.

Supplementary Explanation - Answer 2: Knowledge Base

Question 2 asks for conclusions to be inferred from general knowledge rather than entered directly as facts. A knowledge base contains predicates about restaurants, dishes, and people, together with rules that allow new statements to be derived.

2.1 Meaning and purpose of the general rules

Rule	Plain-language interpretation	Role in the proof
R1	A vegetarian restaurant is a restaurant, and every item it serves is vegetarian food.	Connects the restaurant label to a condition on all served items.
R2	A food item is vegetarian food exactly when it does not contain meat.	Allows movement between the food category and the meat-content property.
R3	Every hamburger contains meat.	Classifies the BigMac as containing meat.
R4	Every salad does not contain meat.	Classifies the SideSalad as not containing meat.
R5	A vegetarian can eat vegetarian food.	Connects Mary and the SideSalad to CanEat.
R6	If a restaurant serves an item that a person can eat, that person can eat at the restaurant.	Derives CanEatAt from serving and food compatibility.

R1 is a one-way rule in this knowledge base. It allows the conclusion that a vegetarian restaurant has the stated serving property. It does not, by itself, allow the reverse conclusion that every restaurant satisfying the serving property is a vegetarian restaurant. Question 4 examines this limitation in more detail.

2.2 Specific facts used as starting points

Fact	Statement	How it is used
F1	Restaurant(McDonalds)	Identifies McDonalds as a restaurant.
F2	Serves(McDonalds, BigMac)	Provides a served item that will be the counterexample.
F3	Hamburger(BigMac)	Activates the hamburger rule R3.
F4	Serves(McDonalds, SideSalad)	Shows that McDonalds also serves the salad.
F5	Salad(SideSalad)	Activates the salad rule R4.
F6	Vegetarian(Mary)	Provides a specific vegetarian who can witness the existential claim.

The facts describe the particular example. The rules are more general: they apply to any entity that satisfies their conditions. Combining the two is what makes the reasoning an inference instead of a restatement of the requested conclusions.

2.3 Inference operations appearing in the proof

Operation	Purpose
Universal instantiation	Apply a universally quantified rule to a particular object, such as BigMac or SideSalad.
Modus ponens	From P and P ⇒ Q, derive Q.
Conjunction introduction	Combine two established statements P and Q into P ∧ Q.
Counterexample to a universal	One served item that is not vegetarian food makes the claim that all served items are vegetarian food false.
Modus tollens	From P ⇒ Q and ¬Q, derive ¬P.
Existential generalisation	From a particular witness, such as Mary, infer that at least one such person exists.

Supplementary Explanation - Answer 2: Expanded Inference Steps

2.4 Deriving that McDonalds is not a vegetarian restaurant

The key idea is to find one item served by McDonalds that is not vegetarian food. The BigMac provides that counterexample. Since R1 requires every served item of a vegetarian restaurant to be vegetarian food, a single violating item is enough to reject the vegetarian-restaurant classification.

Step	Derived statement	Reason and interpretation
1	Hamburger(BigMac)	Given fact F3. The item BigMac is classified as a hamburger.
2	ContainsMeat(BigMac)	Apply R3 to the hamburger fact.
3	¬VegetarianFood(BigMac)	Use R2: an item containing meat cannot be vegetarian food.
4	Serves(McDonalds, BigMac)	Given fact F2. McDonalds serves the item identified above.
5	Serves(McDonalds, BigMac) ∧ ¬VegetarianFood(BigMac)	Combine steps 3 and 4. This gives a concrete counterexample.
6	¬∀y (Serves(McDonalds,y) ⇒ VegetarianFood(y))	The universal condition is false because BigMac is served and is not vegetarian food.
7	VegetarianRestaurant(McDonalds) ⇒ ∀y (Serves(McDonalds,y) ⇒ VegetarianFood(y))	Instantiate R1 for McDonalds and take the required implication.
8	¬VegetarianRestaurant(McDonalds)	Apply modus tollens to steps 6 and 7.

The proof does not say that McDonalds serves no vegetarian food. The SideSalad fact shows that it does serve a salad. The conclusion is narrower: it cannot satisfy the strict condition that all items it serves are vegetarian food, because it also serves the BigMac.

2.5 Deriving that some vegetarians can eat at McDonalds

For the second conclusion, the proof uses the SideSalad and Mary. It first establishes that the salad is vegetarian food, then that Mary can eat it, and finally that she can eat at a restaurant serving it.

Step	Derived statement	Reason and interpretation
1	Salad(SideSalad)	Given fact F5.
2	¬ContainsMeat(SideSalad)	Apply R4 to the salad fact.
3	VegetarianFood(SideSalad)	Use the reverse direction of the equivalence R2.
4	Vegetarian(Mary) ∧ VegetarianFood(SideSalad)	Combine F6 with step 3.
5	CanEat(Mary, SideSalad)	Apply R5: a vegetarian can eat vegetarian food.
6	Serves(McDonalds, SideSalad) ∧ CanEat(Mary, SideSalad)	Combine F4 with step 5.
7	CanEatAt(Mary, McDonalds)	Apply R6 to step 6.
8	∃x (Vegetarian(x) ∧ CanEatAt(x, McDonalds))	Use Mary as the witness for existential generalisation.

This conclusion is existential. It claims that at least one vegetarian can eat at McDonalds; Mary is the witness in this knowledge base. It does not claim that every vegetarian can eat there, nor that every item served there is vegetarian food.

Supplementary Explanation - Answer 3: Temporal Relationships

The two sentences in Question 3 contain the same two events: Mary's flight departs, and the speaker eats lunch. Their difference is the time relation between those events. Event predicates identify what happened, while interval predicates represent when it happened.

3.1 Notation used for events and time

Symbol	Meaning
e1, e2	Event variables for the departure and eating events.
f, l	The flight and the lunch involved in the events.
i, j	Time intervals associated with the departure and eating events.
Flight(f), FlightOf(f,Mary)	f is a flight belonging to Mary.
Departing(e1), Departer(e1,f)	e1 is a departure event and its departing flight is f.
Eating(e2), Eater(e2,Speaker), Eaten(e2,l)	e2 is the speaker's event of eating lunch l.
IntervalOf(e,i)	Event e is associated with interval i.
Precedes(i,j)	Interval i ends before interval j begins.
Overlaps(i,j)	The assignment uses this relation for the simultaneous/coincident timing of the two events.
Now	The time at which the sentence is spoken.

The formulas use existentially quantified event and interval variables because the sentences describe particular events. They also state that the departure occurs before Now, making it a past event. The main contrast is the relation between i and j.

3.2 Sentence (a): "When Mary's flight departed, I ate lunch."

The assignment represents this sentence with Overlaps(i,j). The intended reading is that the speaker's lunch-eating interval coincides with the interval of the flight departure. Both intervals are before the speaking time:

$$IntervalOf(e1,i) \wedge IntervalOf(e2,j) \wedge Overlaps(i,j) \wedge Precedes(i,Now) \wedge Precedes(j,Now)$$

In this representation, the simple past "ate" places the eating event at the reference time associated with the departure. The speaker was eating lunch at the time Mary's flight departed.

Timeline for sentence (a)

Event	Past interval	Relation
Mary's flight departure (i)	[==========]	
Speaker eats lunch (j)	[==========]	The two intervals coincide in the assignment's representation.
Speaking time (Now)	Now	Both events are before the speaking time.

3.3 Sentence (b): "When Mary's flight departed, I had eaten lunch."

The past perfect "had eaten" places the eating interval j before the departure interval i. The assignment therefore uses Precedes(j,i), while the departure interval remains before Now:

$$Precedes(j,i) \wedge Precedes(i,Now)$$

Timeline for sentence (b)

Event	Past interval	Relation
Speaker eats lunch (j)	[==========]	
Mary's flight departure (i)	[==========]	j precedes i.
Speaking time (Now)	Now	Both events are before the speaking time.

Supplementary Explanation - Answer 4: Defining a Vegetarian Restaurant

Question 4 asks why the proposed rule is not a satisfactory definition and how it can be improved. The issue is not merely the choice of symbols: a definition must express both the required conditions and the direction in which the inference is allowed.

4.1 The original rule and its limitations

Original rule: $\forall x\ (VegetarianRestaurant(x) \Rightarrow Serves(x, VegetarianFood))$

Limitation	Why it matters
One-way implication	The rule only tells us what follows from already knowing that x is a vegetarian restaurant. It does not define when an ordinary restaurant qualifies.
Too weak a serving condition	A restaurant serving one vegetarian dish could satisfy the stated condition even if it also serves meat.
Food category is not expressed per dish	Using VegetarianFood as one object does not identify each particular item served by the restaurant.
Restaurant status is missing	The rule does not explicitly require x to be a restaurant.

The McDonalds example illustrates the second limitation. McDonalds serves SideSalad, but it also serves BigMac. A definition that requires merely one vegetarian item would not distinguish this restaurant from a restaurant that serves only vegetarian items.

4.2 A stronger definition

The assignment proposes that a vegetarian restaurant must be a restaurant, must serve at least one item, and must serve only vegetarian food:

$$\forall x\ (VegetarianRestaurant(x) \Leftrightarrow (Restaurant(x) \wedge \exists y\ Serves(x,y) \wedge \forall z\ (Serves(x,z) \Rightarrow VegetarianFood(z))))$$

The variable z is used in the universal condition to make the two scopes easy to read. This is the same intended condition as the assignment's formula: there is at least one item served, and every item served is vegetarian food.

4.3 What each part contributes

Part of definition	Purpose
Restaurant(x)	Excludes entities that are not restaurants.
∃y Serves(x,y)	Requires the restaurant to serve at least one item, avoiding an empty-menu case.
∀z (Serves(x,z) ⇒ VegetarianFood(z))	Requires every served item to be vegetarian food. One non-vegetarian item makes this condition false.
⇔	Makes the condition a two-way definition: the properties are both necessary and sufficient.

4.4 Testing the definition against examples

Example	Result under the improved definition	Reason
McDonalds in the knowledge base	Not vegetarian restaurant	It serves BigMac, which is not vegetarian food, so the universal condition fails.
A restaurant serving one or more dishes, all vegetarian	Qualifies, provided it is a restaurant	It satisfies the restaurant condition, the non-empty serving condition, and the all-items condition.
A restaurant with no served items	Does not qualify	The existential condition ∃y Serves(x,y) is false.
A non-restaurant that serves vegetarian items	Does not qualify	Restaurant(x) is false, even if the food condition holds.

Conclusion

The four questions demonstrate four related uses of formal logic in natural language processing. FOL translations make quantifier scope explicit; a knowledge base derives conclusions from general rules and particular facts; temporal predicates distinguish simultaneous events from events that occur earlier; and a careful definition states the necessary and sufficient conditions for category membership. The added explanations follow the predicates, rules, and examples used in the original assignment.